Title:

AI-Assisted sample preparation approach to improve throughput and reproducibility in metabolomics.

Presenter:

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AI-Assisted Metabolomics Sample Preparation Enhances Analytical Accuracy and Result Reliability

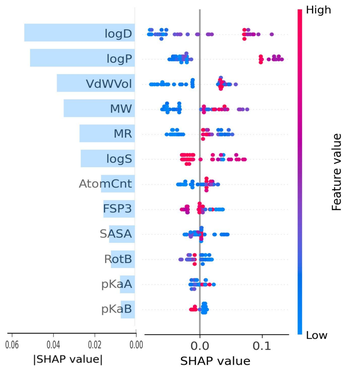
BACKGROUND

In metabolomics, drying is commonly used during sample preparation to support long-term storage and increase analyte concentration. However, drying can selectively deplete metabolites and lead to incomplete reconstitution, introducing systematic bias and compromising analytical accuracy. Here, we use AI to pinpoint the metabolites most affected by drying and connect their vulnerability to underlying chemical features to provide mechanistic insight. This framework informs more robust sample-prep design and improves throughput and reproducibility in metabolomics.

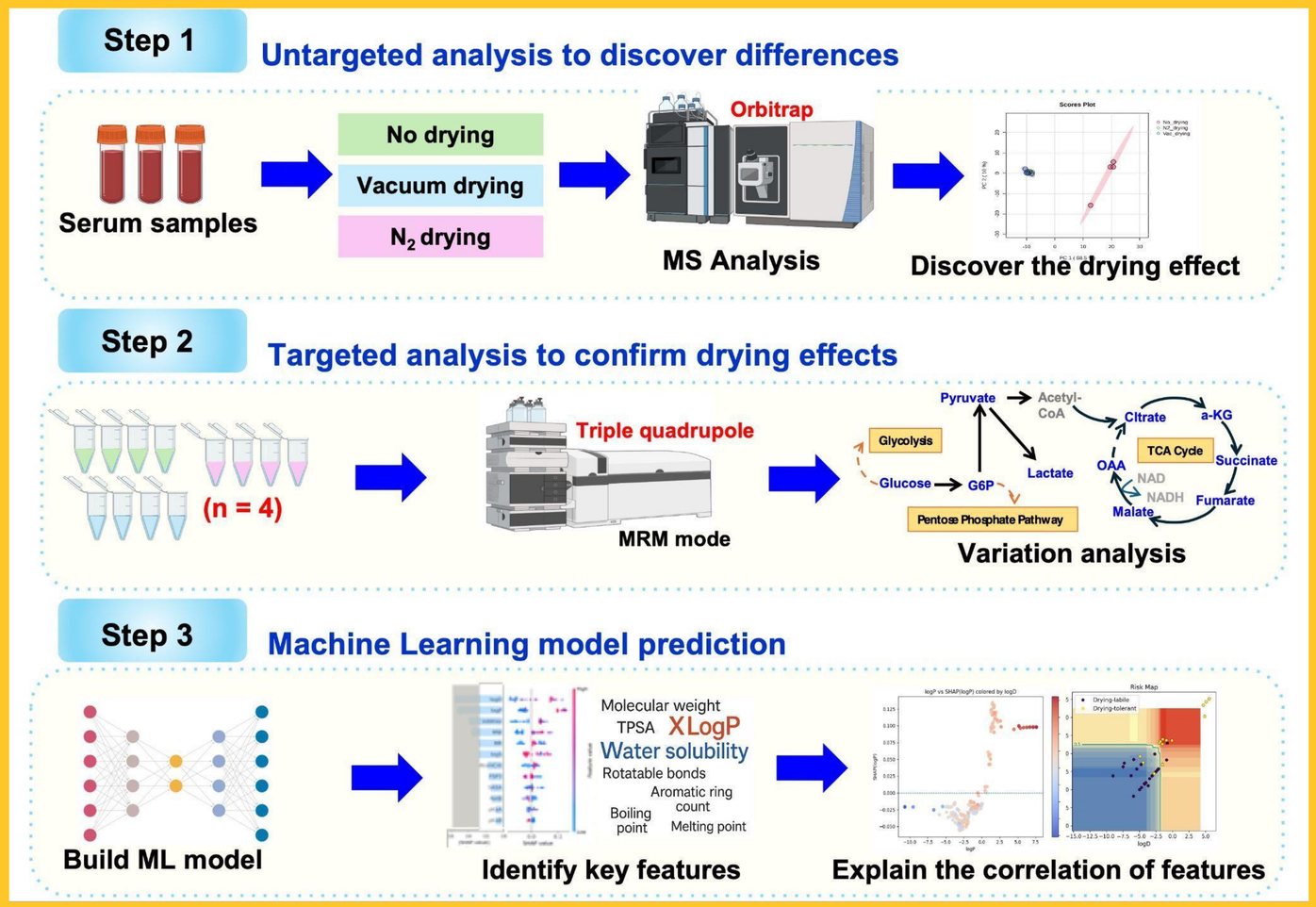
METHODS

1. Samples & prep: Serum aliquots.
2. Untargeted analysis: High-resolution LC–MS (Orbitrap).
3. Targeted validation: LC–MS/MS (Triple quadrupole, MRM mode).
4. AI / data analysis: Python 3.12 for modeling and interpretation; Chemaxon for molecular descriptors; MetaboAnalyst 6.0 for statistics/visualization.

RESULTS



AI ranks the most important chemical properties predicting drying sensitivity.



Workflow overview. Serum samples were prepared under three conditions (no drying, vacuum drying, and N_2 drying). Untargeted Orbitrap MS was used to detect global drying-related changes, targeted triple-quadrupole LC–MS/MS (MRM; n = 4) validated key effects, and an interpretable machine-learning model identified molecular features and predicted metabolites most sensitive to drying to improve analytical accuracy and reliability.

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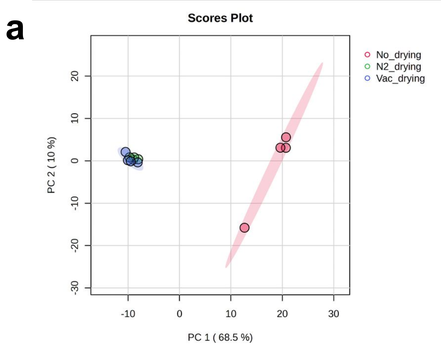
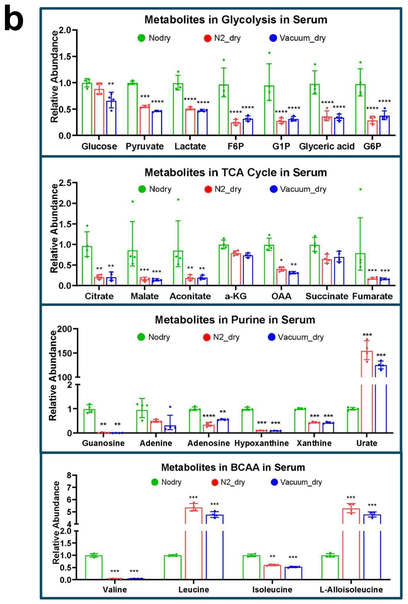
a



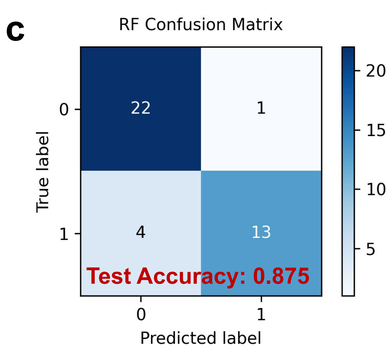
Untargeted PCA analysis shows that drying alters the metabolite profile.

b



Targeted analysis shows that drying significantly impacts key metabolic pathways.

c



The AI model predicts metabolite drying sensitivity with 87.5% accuracy.

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